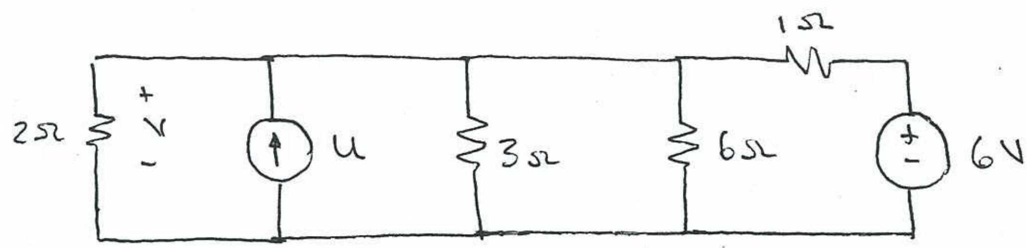
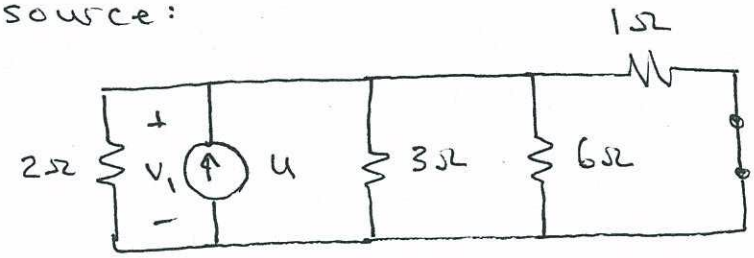
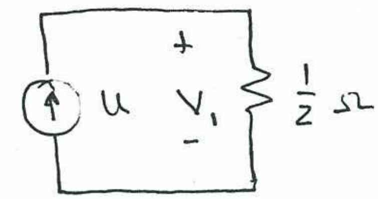
4.1.



Response to current source, u. Kill voltage source:

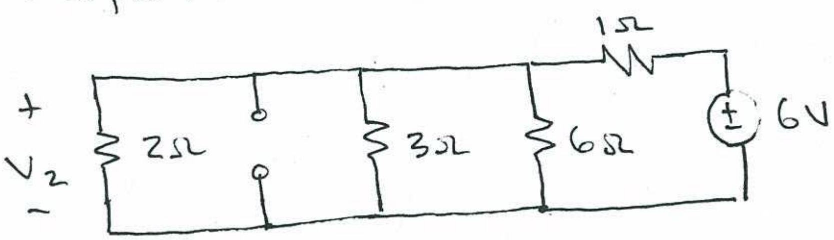


$\Rightarrow$

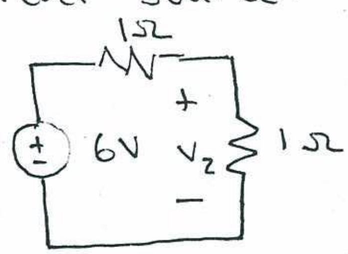


$$V_1 = u\left(\tfrac{1}{2}\,\Omega\right) = \frac{u}{2}\ \text{V}$$

Response to 6V source. Kill current source:



$\Rightarrow$



$$V_2 = 6\text{V}\left(\frac{1\,\Omega}{1\,\Omega + 1\,\Omega}\right) = 3\text{V}$$

Superimpose:

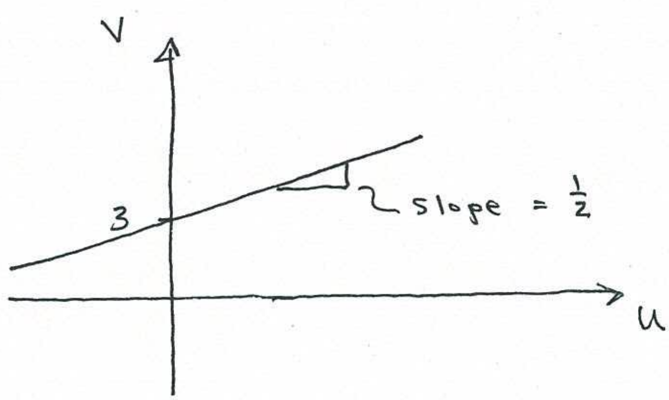
$$V = V_1 + V_2 \quad \Rightarrow \quad \underline{\underline{V = 3\text{V} + \frac{u}{2}}}$$

4.2.

From problem 1:

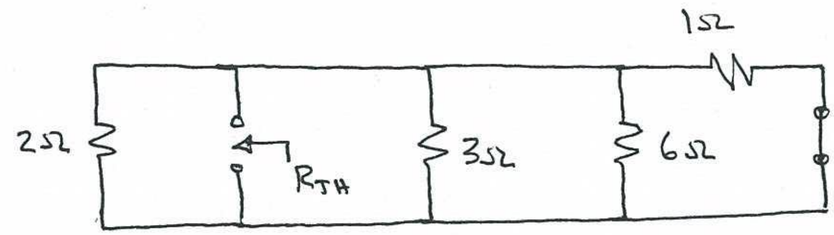
$$V = \frac{u}{2} + 3V$$

This is the input-output relation for the circuit. u is the input, V is the output.



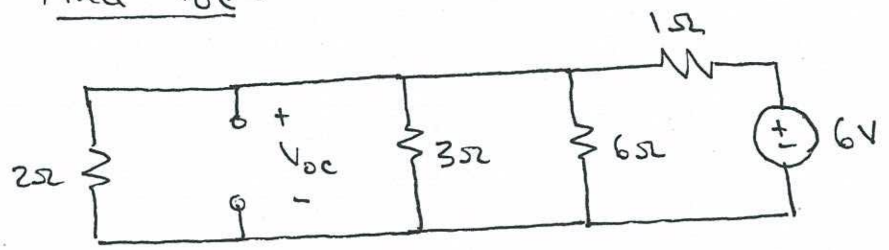
4.3

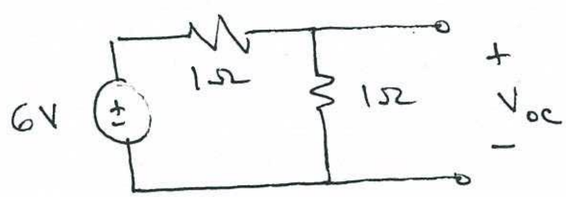
Find R_{TH}:



$$R_{TH} = \frac{1}{\frac{1}{2\Omega} + \frac{1}{3\Omega} + \frac{1}{6\Omega} + \frac{1}{1\Omega}} \Rightarrow \underline{R_{TH} = \frac{1}{2}\,\Omega}$$

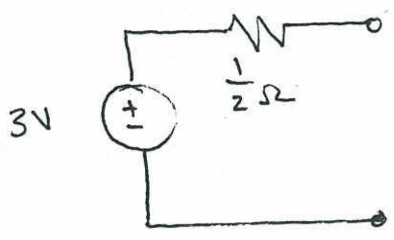
Find V_{oc}:



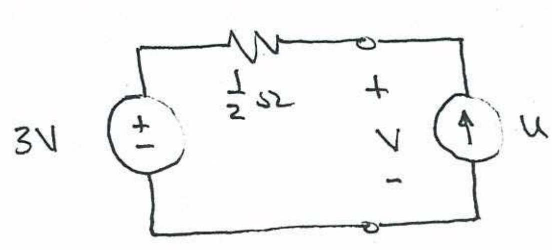


$\Rightarrow$ $\underline{V_{oc} = 3V}$

Thevenin Circuit:

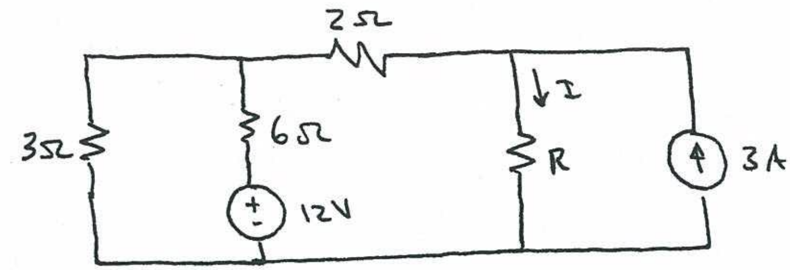


Loaded circuit:

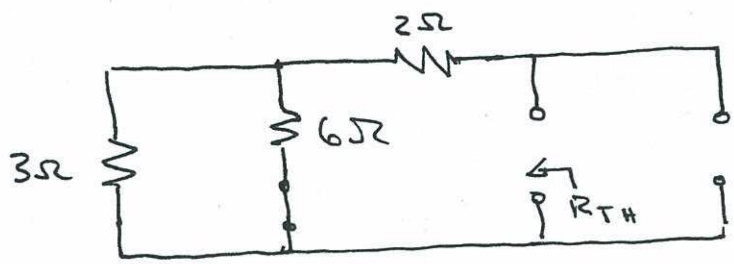


$$\Rightarrow V = \frac{u}{2} + 3\text{ V}$$

4.4



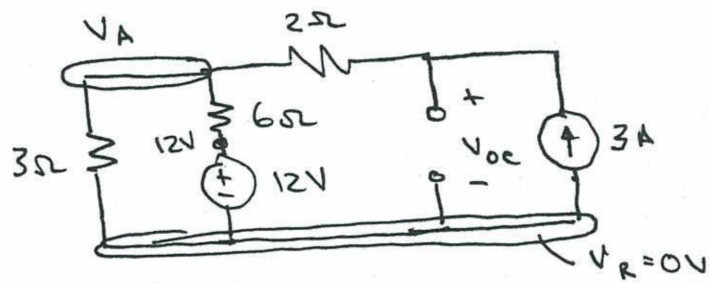
(a) Remove resistor R, kill sources and determine Thevenin resistance:



$$R_{TH} = \frac{(3\Omega)(6\Omega)}{3\Omega + 6\Omega} + 2\Omega$$

$$R_{TH} = 4\Omega$$

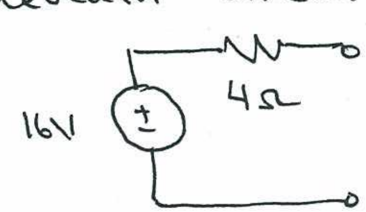
Replace sources & Find V_{oc}:



$$\frac{V_A}{3\Omega} + \frac{V_A - 12V}{6\Omega} - 3A = 0$$

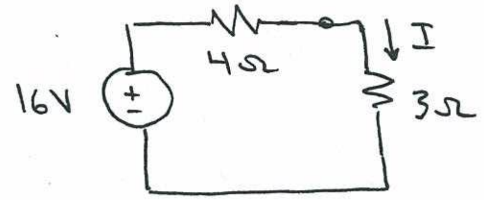
$$V_A = 10V \Rightarrow V_{oc} = 16V$$

Thevenin Circuit:



4.4 (cont'd)

(b) Replace resistor & set $R = 3\Omega$:

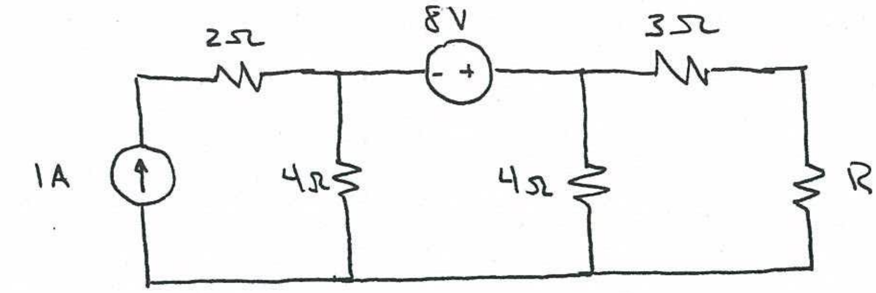


$I = \frac{16V}{7\Omega} \Rightarrow \boxed{I = \frac{16}{7}\,A}$

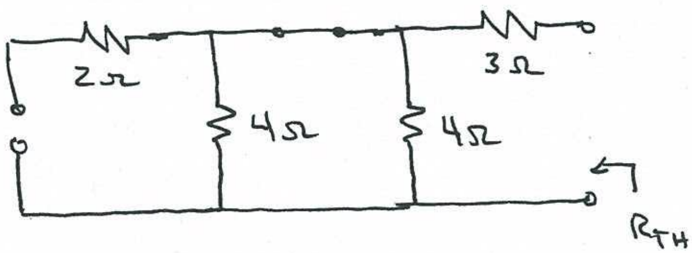
(c) Maximum power when $R = R_{TH}$:

$\boxed{R = 4\Omega}$

4.5



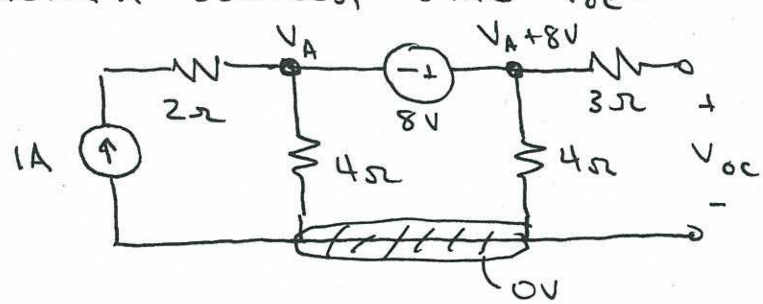
(a) Kill sources, find resistance "seen" by R:



$$R_{TH} = 3\Omega + \frac{(4\Omega)(4\Omega)}{4\Omega + 4\Omega}$$

$$\underline{R_{TH} = 5\Omega}$$

Return sources, find V_{oc}:

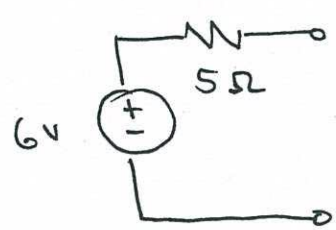


$$-1A + \frac{V_A}{4\Omega} + \frac{V_A + 8V}{4\Omega} = 0$$

$$V_A = -2V$$

$$V_{oc} = V_A + 8V = \underline{\underline{6V}}$$

Thevenin circuit:

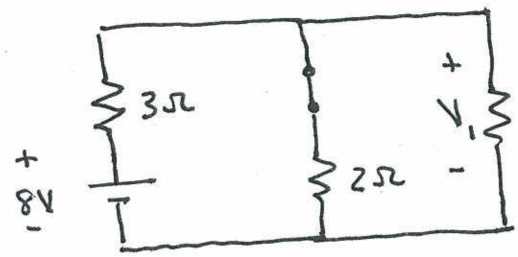


(b) Maximum power transfer when $R = R_{TH}$:

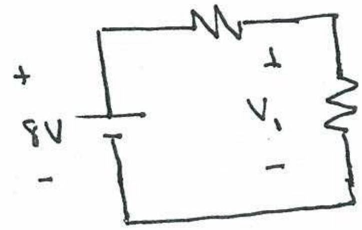
$$\boxed{R = 5\Omega}$$

4.6

Contribution from 8V source:



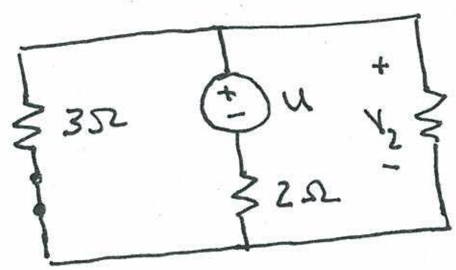
$\Rightarrow$



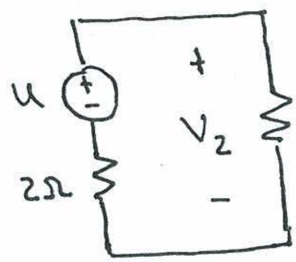
$1.5\,\Omega = \dfrac{6\cdot 2}{6+2}$

Voltage divider: $V_1 = \dfrac{1.5\,\Omega}{3\,\Omega + 1.5\,\Omega} \cdot 8V = \left(\frac{1}{3}\right) 8V \Rightarrow \underline{\underline{V_1 = \frac{8}{3} V}}$

Contribution from "u" Volt input:



$\Rightarrow$



$2\,\Omega = \dfrac{(6\,\Omega)(3\,\Omega)}{6\,\Omega + 3\,\Omega}$

Voltage divider: $V_2 = \left(\dfrac{2\,\Omega}{2\,\Omega + 2\,\Omega}\right) u \Rightarrow \underline{\underline{V_2 = \frac{u}{2} V}}$

Superimpose:

$V = V_1 + V_2 = \left(\frac{8}{3} V\right) + \left(\frac{u}{2}\right) V \Rightarrow \boxed{V = \frac{8}{3} + \frac{u}{2}}$

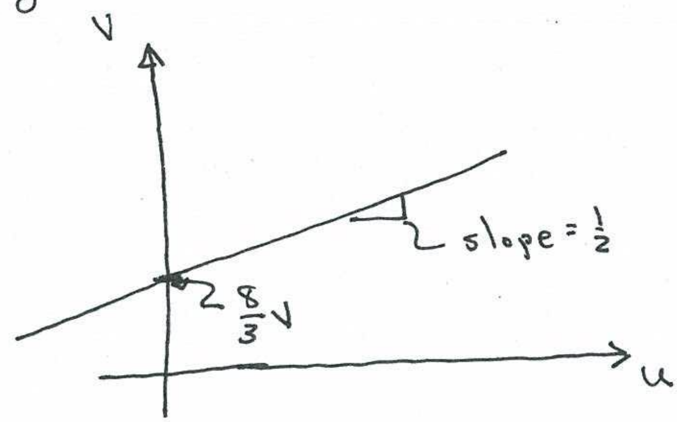
4.7

Using superposition (in previous problem), we found that:

$$\boxed{V = \frac{8}{3} + \frac{1}{2} u}$$

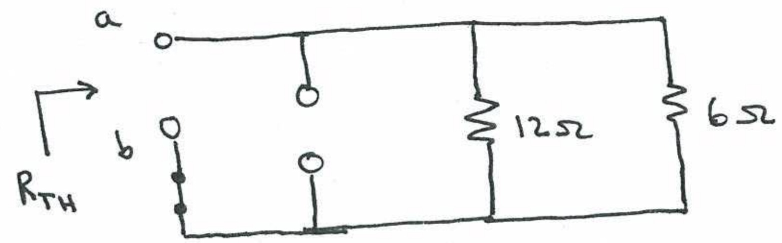
This is a valid input-output relationship!

Plotting:



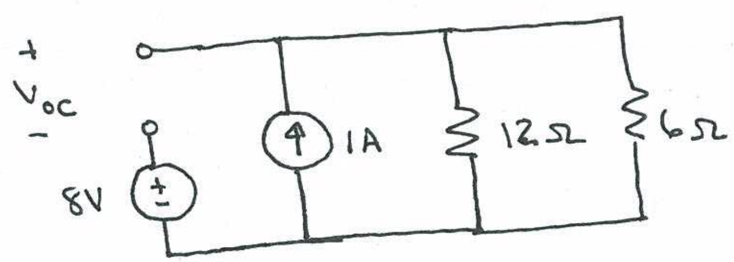
4.8

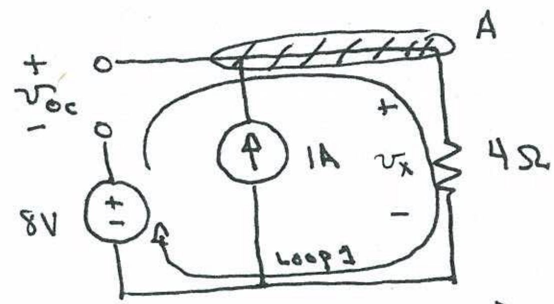
Find R_{TH}:



$$\Rightarrow R_{TH} = \frac{12 \cdot 6}{12+6} = \underline{\underline{4\Omega}}$$

Find V_{oc}:



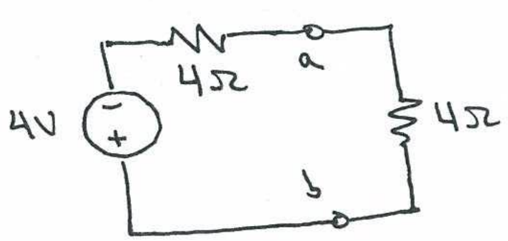


equivalent circuit.

KCL at "A" + Ohm's Law $\Rightarrow v_x = (4\Omega)(1A) = 4V$

KVL, Loop 1: $v_{oc} = v_x - 8V = -4V$

Thevenin Circuit:

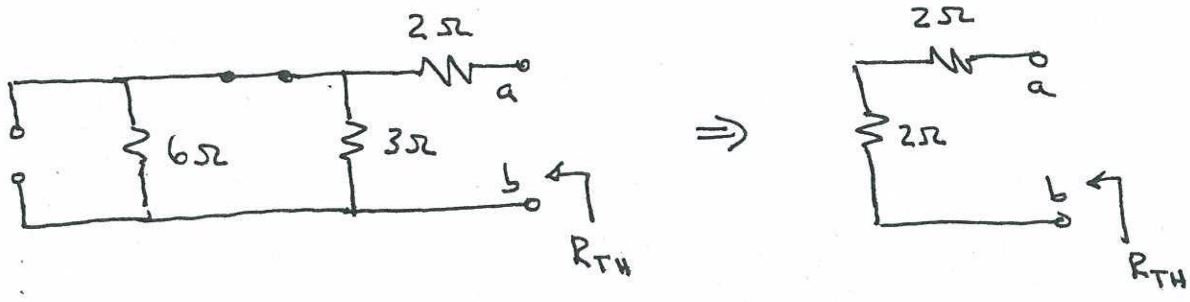


$$\Rightarrow v_{ab} = -4V\left(\frac{4\Omega}{4\Omega + 4\Omega}\right)$$

$$\boxed{v_{ab} = -2V}$$

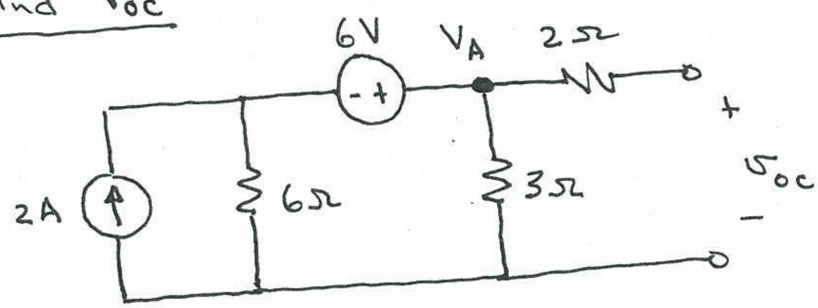
4.9

Find R_{TH}:



$$R_{TH} = 2\Omega + 2\Omega = \underline{\underline{4\Omega}}$$

Find V_{oc}:

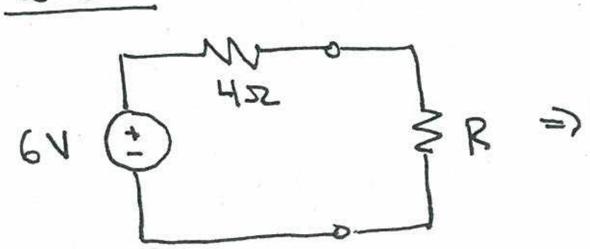


Nodal analysis to find V_A:

$$-2 + \frac{V_A - 6}{6\Omega} + \frac{V_A}{3\Omega} = 0 \Rightarrow 3V_A = 18 \Rightarrow V_A = 6V$$

$$v_{oc} = V_A = \underline{\underline{6V}}$$

Thevenin Circuit



⟹ Maximum power when

$$\boxed{R = R_{TH} = 4\Omega}$$